3	**(a)**	$F = q E = (2.0 \text{ x } 10^{-9})(1.5 \times 10^{5})$ $F = 3.0 \text{ x } 10^{-4}$ N direction: upwards	M1 A1 B1
	(b)	Loss in gravitational potential energy $= m\, g\, (h + x)$ Gain in elastic potential energy and electric potential energy $= ½\, k\, x^2 + F_E\, (h + x)$	 C1 C1

		By conservation of energy, $m\,g\,(h + x) = \frac{1}{2}\,k\,x^2 + F_E\,(h + x)$ $(1.0 \times 10^{-4})(9.81)(0.515) = \frac{1}{2}\,k\,(0.015)^2 + (3.0 \times 10^{-4})(0.515)$ $k = 3.12\ \text{N m}^{-1}$ (1 mark if overall correct method to solve using conservation of energy.)	A1

4	(a)	(Simple harmonic motion is the motion of a particle about a fixed point such that	